Mohit Sharma

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PROFESSIONAL SUMMARY

AI Engineer specializing in LLM alignment (RLHF) and real-time computer vision systems, building scalable low-latency AI pipelines.
Delivered production-grade solutions processing 5,000+ images/day with sub-120ms latency.
Expertise in PyTorch, FastAPI, and GPU optimization, focused on performance, reliability, and real-world impact.

SKILLS

Languages: Python, C++ | ML/DL: PyTorch, TensorFlow, Scikit-learn, Transformers | GenAI: RLHF, LLM Pipelines, Prompt Engineering, RAG (Retrieval-Augmented Generation)
Computer Vision: OpenCV, Segmentation, Object Detection, Mask R-CNN | MLOps: Docker, FastAPI, CI/CD | Optimization: CUDA, ONNX, TensorRT | Tools: Git, Linux

EXPERIENCE

AI Developer – Sodio Technologies | Remote – Full-Time | June 2026 - Present

- Built a post recommendation system to deliver personalized content recommendations using user and content signals.
- Developed AI agents to automate health-data processing and workflows for data collected from iPhone/Apple Health.
- Built autonomous AI testing agents to automate software testing workflows and reduce manual testing effort.
- Designed and integrated AI-powered pipelines and backend services for production applications.
- Worked across LLMs, agentic workflows, APIs, and AI automation to deliver scalable production features.
- Worked directly with clients to gather requirements, present AI solutions, incorporate feedback, and drive features from initial discussion through implementation

Data Scientist – Hairscope | Remote – Full-Time | July 2025 - June 2026

- Built scalable FastAPI backend processing 5,000+ images/day with <120ms latency for real-time inference
- Developed and deployed deep learning models using PyTorch achieving >90% accuracy on production datasets
- Designed end-to-end computer vision pipelines including preprocessing, training, and inference
- Optimized GPU training workflows, reducing training time by ~20% and improving system scalability
- Integrated AI models into production systems ensuring high availability and performance
- Improved inference efficiency through model optimization and batch processing techniques

AI Engineer (Junior) – Centilytics | Bangalore – Full-Time | Jan 2025 - July 2025

- Built transformer-based NLP pipelines for text classification and information extraction tasks
- Fine-tuned deep learning models to improve accuracy and domain-specific performance
- Integrated ML models into scalable backend systems enabling real-time inference
- Optimized model performance through debugging, hyperparameter tuning, and system-level improvements
- Designed data preprocessing pipelines for handling large-scale structured and unstructured datasets
- Collaborated with engineering teams to deploy robust and maintainable AI solutions

Junior Developer Intern – Hairscope | Internship | July 2024 – Dec 2024

- Developed computer vision pipelines using OpenCV and NumPy
- Built ML models for classification and detection tasks
- Assisted in deploying AI solutions into production systems

PROJECTS

AI Autonomous Testing Agents

Agentic AI system developed at Sodio Technologies to automate software testing workflows.

- Built autonomous AI agents to execute and automate software testing workflows
- Designed agent-driven automation to reduce manual testing effort and improve testing coverage

Post Recommendation System

Recommendation system developed at Sodio Technologies for personalized post discovery.

- Designed a recommendation pipeline using user and content signals to support personalized content ranking
- Worked on recommendation logic and AI-driven ranking workflows for production use

Driver Drowsiness Detection System

Real-time computer vision system to detect driver fatigue and prevent accidents using edge devices.

- Built CNN-based pipeline using Python, OpenCV, and PyTorch for real-time face and eye tracking
- Deployed on Raspberry Pi achieving <150ms latency with continuous monitoring and alert system
- Registered copyright with Government of India (SW-19841/2024), demonstrating originality

Hair Strand Detection (Instance Segmentation)

Computer vision system for precise hair strand segmentation and analysis using deep learning.

- Built Mask R-CNN (ResNet) model achieving 0.84 mAP and 0.78 IoU on custom dataset
- Designed preprocessing and augmentation pipelines to improve accuracy and robustness
- Optimized inference using ONNX/TensorRT for efficient production deployment

AI Dermatology & Hair Analysis Platform

AI-powered diagnostic platform for skin and hair condition analysis using deep learning models.

- Developed end-to-end system using deep learning and foundation model embeddings (95% accuracy)
- Built scalable FastAPI backend for real-time inference and integrated Streamlit-based UI
- Optimized model performance and deployment pipeline for production-level usage

EDUCATION

M.Tech (Distance Learning) Computer Science RGPV (2025–2027) Bhopal, India
B.Tech CGPA: 7.6 AI & Data Science RGPV (2021-2025) Bhopal, India